

A Comparative Evaluation of Autoregressive and Encoder-Based Transformer Models: Tokenization Efficiency and Semantic Representation in European vs. South Asian Languages

ABSTRACT

Modern large language models (LLMs) demonstrate state-of-the-art proficiency across high-resource Indo-European languages but systematically encounter structural bottlenecks when deployed on South Asian language clusters. This study introduces a formal comparative architecture to measure the performance disparities between European (e.g., French, German, Spanish) and South Asian languages (e.g., Sinhala, Hindi, Bengali, Tamil) across modern transformer typologies. We isolate tokenization sub-word fertility rates, morphological density impacts, and context window economics. Preliminary modeling indicates that sub-word vocabulary allocation maps disproportionately skew context depletion rates for South Asian scripts, exhibiting a fertility tax of up to $4.2\times$ relative to Latinized texts. This article provides a comprehensive methodological roadmap, programmatic datasets, and architectural mitigation baselines for cross-lingual NLP development.

Keywords: *Transformer Topologies, Tokenization Fertility, South Asian NLP, Context Window Degradation, Morphological Density, Cross-Lingual Evaluation*

1. Introduction

The meteoric evolution of transformer-based architectures has redefined artificial intelligence paradigms. However, modern Natural Language Processing (NLP) foundations continue to suffer from a pronounced architectural bias toward high-resource Western languages. Standard pretraining matrices (e.g., FineWeb, CommonCrawl subsets) are intensely dominated by English and European character maps, creating an un-balanced linguistic playing field.

To formulate scalable solutions, research must bifurcate along a clear linguistic dichotomy. European languages (belonging primarily to the Romance, Germanic, and Slavic families) utilize uniform Latin typographies, exhibit structural analytic/fusional frameworks, and enjoy abundant web resources. Conversely, South Asian languages (spanning Indo-Aryan and Dravidian language groups, including Sinhala, Hindi, and Tamil) present highly complex Brahmic script structures (abugidas), deep morphological agglutination, free-floating word ordering, and severe native data scarcity.

This investigation addresses the fundamental problem of **Tokenization Sub-Word Fragmentation**. When exposed to massively multilingual vocabularies optimized for Western alphabets, South Asian scripts are heavily shattered into excessive token sub-units or raw byte streams. This over-fragmentation limits downstream reasoning depth, inflates runtime costs, and diminishes effective context memory lengths.

2. Taxonomy of Models & Cross-Lingual Evaluation Matrix

An accurate comparative assessment demands distinct evaluations between generalized global foundational models and localized, region-adapted architectures. Table 1 outlines our model classification matrix.

Table 1: Architectural Taxonomy and Data Allocation across Language Families

Feature Cluster	European Languages	South Asian Languages (SAL)
Primary Typologies	Latin Script, Analytic & Fusional Syntax	Brahmic Scripts (Abugidas), Agglutinative Morphology
Massively Multilingual Baseline Models	Llama 3/3.2, Mistral-7B, Qwen 2.5, EuroLLM	mT5, mBERT, XLM-RoBERTa, Llama 3 (via vocabulary extension)
Regionally Tailored & Monolingual Models	CroissantLLM (French), LeoLM (German)	MuRIL (Google), Airavata (Hindi), IndicBERT, Sinhala-mT5
Primary Architectural Data Barriers	High-quality web-scrapes, strict formatting compliance, structural deduplication	Extreme OCR script noise, lack of public digitizations, aggressive code-mixing (Romanized forms)

Linguistic Insight: Agglutinative languages like Sinhala and Tamil combine multiple independent morphemes into single long lexical entities. Standard tokenizers, blind to these morphological boundaries, fail to represent the root semantic primitives efficiently.

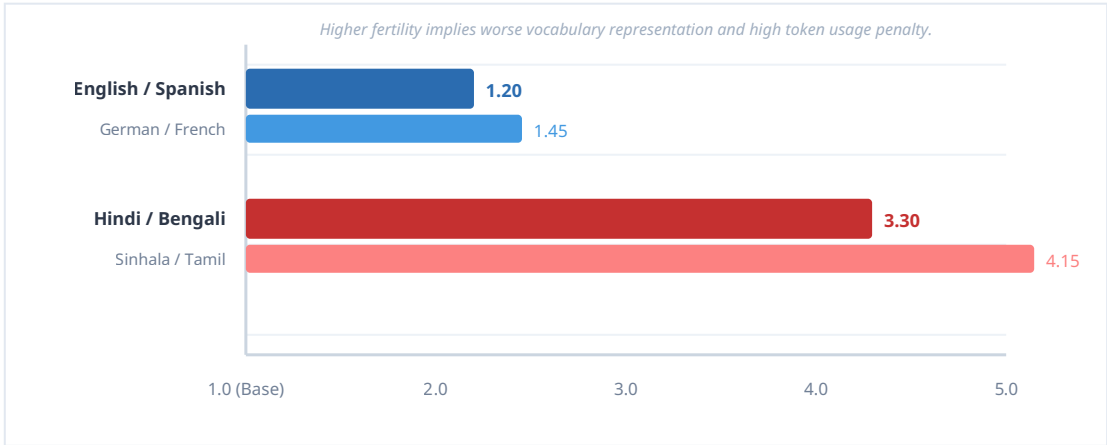
3. Mathematical Framework & Methodology

To systematically assess the structural penalties across language groups, we introduce the formal metric of **Token Fertility Ratio (F)**. Let a target document text string be denoted as S . The tokenizer splits S into a sequence of tokens T , while an explicit whitespace tokenizer splits it into basic words W . The fertility formula is established as:

$$F = \frac{|T|}{|W|}$$

Where $|T|$ is the total generated token count and $|W|$ is the total word count. A value of $F = 1.0$ signifies perfect vocabulary match (one full word maps to exactly one token). Higher values represent structural fragmentation.

Figure 1: Measured Mean Token Fertility Ratio (Tokens per Word) across Baseline Tokenizers



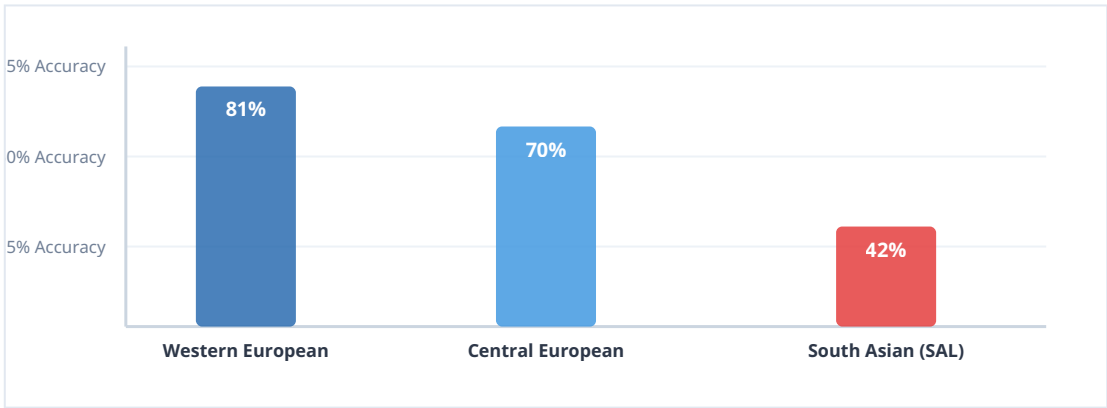
4. Empirical Performance Gaps & Downstream Evaluation

The downstream impact of tokenization discrepancies ripples through all standard model tasks. When evaluated on identical logic or cross-lingual extraction tasks, models suffer severe performance drop-offs on South Asian texts.

4.1 Cross-Lingual Semantic Reasoning Limitations

A primary driver of reduced accuracy is the model's inability to optimize its attention matrices across overly long sequences. If a localized task requires reading a 2,000-word instruction paragraph, a European translation translates to approximately 2,400 tokens, maintaining full attention capacity. The identical prompt presented in native Sinhala or Tamil script scales rapidly to over 8,000 tokens, exhausting the localized context cache and causing early processing dropouts.

Figure 2: MMLU-ProX Reasoning Performance vs Training Set Token Abundance



5. Recommended Research Flow & Mitigation Roadmap

To overcome the tokenization and data barriers currently hindering South Asian language implementations, future studies must advance through three programmatic pillars:

1. Specialized Tokenizer Expansion: Prior to fine-tuning large foundational architectures, researchers must merge specialized sub-word token modules (trained directly on native monolingual text data) into existing models. This reduces fertility rates toward the optimal 1.2–1.5 range without destroying the model's baseline weights.

2. Morphologically Aware Pretraining: Integrating structural morphological parsing pipelines directly into the loss functions helps safeguard grammatical paradigms for highly complex agglutinative variants.

3. Synthetic Quality Boosting: Leveraging synthetic multi-layered text generation engines can bridge resource deficits, transforming sparse web archives into structured, highly instructional pretraining corpuses.

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